



Electrical and optical performances investigation of planar solar blind photodetector based on IZTO/Ga₂O₃ Schottky diode via TCAD simulation

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Abstract

The development of sophisticated solar-blind photodetector devices is being motivated by the increasing need for solar-blind sensors with remarkable photosensitive qualities. This work is a modeling and simulation of electrical and optical properties of a Schottky photodetector based on gallium oxide (Ga₂O₃), one of the most promising wide-band-gap material. It focuses on the optimization of the device structure design, assumed fully transparent to visible light. In addition, analysis of electrical and optical properties of β -Ga₂O₃-based photodetector was carried out to accurately describe physical phenomenon through the semiconductor bulk. The impact of several parameters of the device on its performances is such as layer thickness, doping concentration, and electron affinity, anode work function and different cathode contact materials. Besides that, and for more understanding the real behaviors within different layers of the photodetector, diverse optical parameters were varied, for instance the light power intensity and the spectral response by changing the optical wavelength. The obtained results show a promising performance enhancement compared to previous works, for instance the photo to dark current ratio of 3×10^4 , the good spectral responsivity in UV light illumination [250–350 nm], in addition to the relatively high detectivity of 2.5×10^9 (Jones).

Keywords Gallium oxide · Photodetector · Silvaco · UV light · Wide bandgap

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1 Introduction

There are numerous wavebands within the electromagnetic spectrum of ultraviolet (UV) radiation, such as UVA, UVB, UVC, and vacuum UV (VUV) (Hou et al. 2020). The ozone layer mostly absorbs UVC and VUV light from solar UV radiation before it reaches the earth's surface, where UV wavelengths range of 200–280 nm is known as the "solar-blind" waveband (Lee et al. 2017).

The rapid development in power electronic, automotive and optoelectronic components are mainly driven by advances in specific wide-band-gap semiconductor-based devices with high performance (Shur 2019; Tang et al. 2021). In the recent years, intensive work was devoted to ultra-wide bandgap (UWBG) semiconductors materials including aluminum gallium nitride alloys ($\text{Al}_x\text{Ga}_{1-x}\text{N}$), GaN, diamond, SiC, and a number of other UWBG binary and ternary oxides owing to their outstanding radiation hardness, chemical and thermal stability in addition to an efficient absorption of a specific wavelength light (Qin et al. 2021; Raj et al. 2021), due to the strong inherent critical breakdown electric field, their thin drift region and low on-state resistance (Zhang et al. 2022). Among them, Gallium oxide (Ga_2O_3) is one of the aforementioned semiconductors with an indirect band-gap of 4.5–5.2 eV. Although; it has six polymorphs (α , β , γ , δ , ϵ , and κ) (Galazka et al. 2021; Higashiwaki et al. 2020; Mandal et al. 2022), the beta-phase $\beta\text{-Ga}_2\text{O}_3$ is the most stable, due to its physical properties such as thermal stability for the operational temperature range up to the melting point, a high breakdown electrical field of ~ 8 MV/cm and its high saturation velocity of 1×10^7 cm/s. Furthermore, its important optical properties has demonstrated that is colorless and high transparent to the UV-C range of the light spectrum (Galazka et al. 2021; Labed et al. 2021a). One-dimensional (1D) high surface-to-volume ratio of nanostructured UV detectors has drawn interest due to their high light sensitivity. However, due to the dominance of surface photocurrent and the high temperature of UV detectors based on nanostructured semiconductors, they encounter difficult conditions, and hence, bulk-dominated photocurrents are preferred (Sang et al. 2013).

Light emitting diodes, phototransistors, photo-detectors (PDs) and other optoelectronic devices based on Ga_2O_3 attracted a great interest from the researcher's community in recent years in order to enhance and achieve the desired performance responding to real world issues (Guo et al. 2019; Kumar et al. 2023; Ping et al. 2021). To deal with this concern, and especially to Ga_2O_3 -based PDs, various structures (designs) such as metal–semiconductor–metal (MSM), Schottky barrier diode (SBD) and metal–oxide –semiconductor (MOS) photo-detectors have been proposed. Different elaboration techniques and deposition methods; halide vapor phase epitaxy (HVPE), metal–organic chemical vapor deposition (MOCVD) and molecular beam epitaxial (MBE) are used for the growth of Ga_2O_3 semiconductor layer (Labed et al. 2021b; Patil-Chaudhari et al. 2017; Pearton et al. 2018; Singh Pratiyush et al. 2017; Yadav et al. 2020; Zhang et al. 2022).

Several researchers focused on this field by fabrication of fully transparent solar-blind photodetectors based on $\beta\text{-Ga}_2\text{O}_3$ semiconductor as an active layer. Those works show a high transmittance up to 80% from visible to near infrared (NIR) optical wavelengths, in addition to a low dark current value ($\sim 10^{-9}$ A) (Wei et al. 2014). One of the used structures based on $\beta\text{-Ga}_2\text{O}_3$ Schottky diode was investigated in a recent work (Kim et al. 2022), where both InZnSnO (IZTO) and InSnO (ITO) were used as Schottky and ohmic contacts, respectively. With high rectifying ratio and workfunction of the studied photodetector, it offers several advantages in terms of fast photodetection, responsivity, and precisely it react as self-powered photodetectors (Zou et al. 2018).

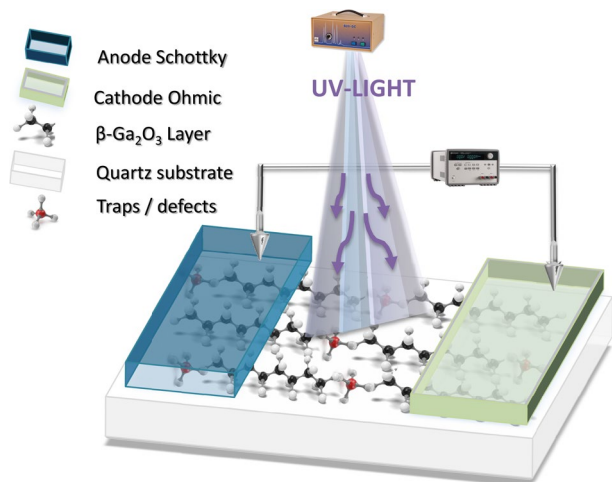
The aim of this work is the analysis and investigation of electrical and optical simulated characteristics of Schottky β -Ga₂O₃-based photodetector (PD). This PD, which consists of a top-contact made of Indium-Zinc-Tin oxide (IZTO) and different transparent conductive oxides (TCOs) (Choi et al. 2010; Heo et al. 2013) as Schottky and ohmic contacts, respectively on β -Ga₂O₃, was analyzed using Silvaco TCAD simulator (Filali et al. 2017). The effect of doping concentration, thickness, workfunction and electron affinity of β -Ga₂O₃ semiconductor were performed. Moreover, the impact of optical parameters of the beam power and wavelength variation on the device performance was carried out to evaluate the figures of merit (FOM) of the PD namely sensitivity, responsivity, and detectivity. A deep analysis of the obtained results could be advantageous for the enhancement of UV self-powered photodetectors.

2 Device design and structure

The simulation study of the Schottky PD based on Ga₂O₃ semiconductor was carried out using Silvaco ATLAS software (Filali et al. 2017), which is a vast environment tool for the design, modelling and simulation of semiconductor-based devices under different conditions by solving the main equations describing conduction in semiconductors such as continuity and Poisson (Labed et al. 2021b; Sze et al. 2021). Such simulation can give an overview of the device functioning and performance. Figure 1 shows a 3D schematic illustration of the structure suggested in this study. In Silvaco ATLAS, each material (layer) in the specified structure is considered as a region with different geometrical and physical parameters. The mesh definition is a crucial parameter in such simulation, which lead to solve the differential equations related to various mechanisms through the material bulk. Mesh refinement is important at junction's location or for metal–semiconductor interfaces.

In this structure; firstly, starting by a quartz substrate with a thickness of 800 μ m which will be taken into account in the simulation stage by enlarging the mesh spacing. Quartz is considered as extremely versatile material used in different optoelectronic domains with outstanding thermal stability properties related to both elaboration or functionality temperature change, excellent optical transmission, effective electrical

Fig. 1 A general 3D schematic view of photodetector based on TCO/Ga₂O₃



insulation and anti-corrosion performances. Secondly, different doping concentration and thicknesses of $\beta\text{-Ga}_2\text{O}_3$ film ranging from 3×10^{16} to $1 \times 10^{18} \text{ cm}^{-3}$ and from 100 to 900 nm, respectively, have been implemented in ATLAS tool. As mentioned before, our structure is top-contacts configuration, where both electrodes IZTO as Schottky contact (150 nm thickness and 1 μm width) and Al as an ohmic contact (150 nm thickness and 1 μm width) were deposited on the top of the effective semiconductor layer ($\beta\text{-Ga}_2\text{O}_3$). In order to elaborate a fully transparent photodetector and to analyze the effect of ohmic contact material on the optical characteristics of the device in question, several transparent conductive oxides (TCOs) such as AZO, ITO, FTO and IZO have been chosen according to their work function values. Other test has been simulated to demonstrate the impact of Schottky contact workfunction on the device performance. Knowing that $\beta\text{-Ga}_2\text{O}_3$ material did not exist in the used software library (Filali et al. 2018; Sze et al. 2021), this issue lead to another important contribution in this study which is expressed by the implementation of different material properties such as the electronic gap, electron affinity, electrons and holes mass and mobility, traps capture cross section and density...etc. After introducing these parameters within the code, it is necessary to input refractive index (n) and extinction coefficient (k) variation with wavelength which are extracted from previous experimental works (Choi et al. 2010; Heo et al. 2013; Labed et al. 2022; Rebien et al. 2002) for the used materials (quartz, contacts and $\beta\text{-Ga}_2\text{O}_3$).

Once the structure has been designed and all the related material's parameters have been implemented, the simulation phase to obtain electrical and optical characteristics of the device in question is initiated under different condition such as beam power intensity, semiconductor affinity and optical wavelength variation. The whole operation was carried out using different models: Shockley Read Hall (SRH), Auger and Band Gap Narrowing (BGN), Arora and impact Selb models to evaluate photodetector performance under dark, illumination and light spectrum variation conditions. Equation 1 describes the recombination SRH rate (R_{SRH}) which is the net recombination of photo-generated carriers inside the semiconductor bulk (Labed et al. 2022; Sze et al. 2021).

$$R_{\text{SRH}} = \frac{pn - n_i^2}{\tau_p \left[n + n_i \exp\left(\frac{E_T}{KT}\right) \right] + \tau_n [p + n_i \exp(\frac{-E_T}{KT})]} \quad (1)$$

where p, n and n_i denotes hole, electron and intrinsic carrier concentration, respectively. τ_p and τ_n are lifetimes for holes and electrons, respectively, T is absolute temperature and K is the Boltzmann constant. In the simulation, we take account the traps in the bandgap of the semiconductor and the values are from (Labed et al. 2022) The difference between trap and intrinsic level energies is denoted by E_T . Table 1 (Pearton et al. 2018; Rebien et al. 2002) lists a set of incorporated $\beta\text{-Ga}_2\text{O}_3$ material properties into the software library.

3 Results and discussion

The obtained simulation results of the pre-designed Schottky $\beta\text{-Ga}_2\text{O}_3$ based photodetector are described below.

Table 1 β -Ga₂O₃ material parameters implemented to software library

Symbol	Parameter	Values
E_g (eV)	Bandgap	4.8
N_D (cm ⁻³)	Donor concentration	3×10^{16}
N_C (cm ⁻³)	Effective density of states in the conduction band	3.7×10^{18}
N_V (cm ⁻³)	Effective density of states in the valance band	5×10^{18}
μ_n (cm ² V ⁻¹ s ⁻¹)	Electron mobility	172
μ_p (cm ² V ⁻¹ s ⁻¹)	Hole mobility	10
m_n	Effective mass of electrons	0.28
m_p	Effective mass of holes	0.35
χ (eV)	Affinity	4
ϵ_r	Relative permittivity	12.6
v_{sat} (cm s ⁻¹)	Saturation velocity (cm s ⁻¹)	10^7

3.1 Dark and illumination I–V characteristics

Dark and illumination currents densities as a function of bias voltage were compared and depicted in Fig. 2a. Ga₂O₃ layer thickness was fixed at 300 nm. Firstly, one can observe a typical rectifying behavior of Schottky structure (IZTO/ β -Ga₂O₃) due to the difference between the work function value of IZTO and the electron affinity of Ga₂O₃ (Lee et al. 2007). Secondly, it can be seen from the graph at reverse bias that under illumination condition (255 nm wavelength and optical power of 100 mW/cm²), an apparent enhancement in the current density compared to that at darkness. Indeed, the reverse current increases from 10⁻³ to 10⁻¹ A when the device is illuminated. In this case, within the depletion region at the β -Ga₂O₃/IZTO interface, the electric field is sufficient to split the photogenerated electron–hole pairs towards their corresponding electrodes, thus creating the photocurrent. Thirdly, at direct bias, a slight shift of the threshold voltage (V_{th}) could be remarked.

As an important figure of merit for such device, the photo to dark current ratio (PDCR) (Young et al. 2008) is defined as the ratio of the current under illumination to the dark current (Eq. 2). Figure 2b shows the dependence of the contrast ratio on the reverse applied voltage under illumination. It is evaluated from 10² to 3 × 10⁴ at -2 and 0 V, respectively.

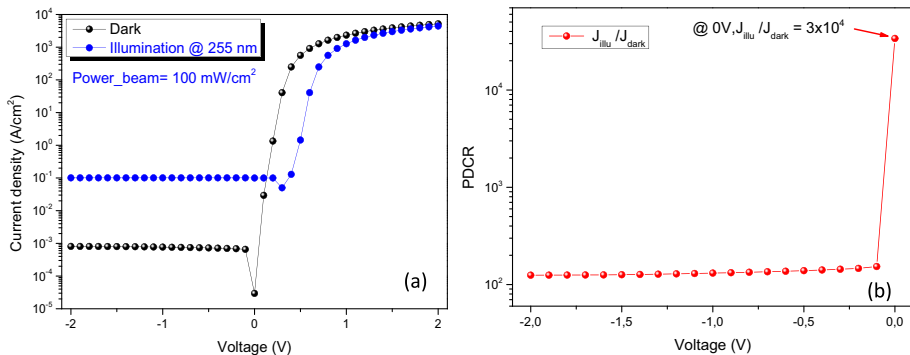


Fig. 2 **a** Simulated I–V characteristics of β -Ga₂O₃ photodiode under darkness and illumination of 100 mW/cm², and **b** calculated contrast ratio as a function bias

Otherwise, it reflect the sensitivity of the device to a specific wavelength (255 nm), this is attributed to the presence of hole-trap states associated with oxygen atoms, which affect the photoresponse mechanism (Labeled et al. 2022).

$$\text{PDCR} = \frac{I_{\text{illum}}}{I_{\text{dark}}} \quad (2)$$

where I_{illum} and I_{dark} are the current under illumination and dark, respectively. Higher sensitivity reflects high and well rectifying behavior of Schottky-based photodetector.

3.2 Effect of light intensity

As presented in Fig. 3, the increase in beam power intensity has a seeming impact on the photocurrent at reverse and 0 V bias, where at reverse voltage $[-2$ to near 0 V] it rise from 8×10^{-4} to 2×10^{-3} A/cm² for 1 and 10^3 mW/cm² beam intensities, respectively. At 0 V, the photocurrent density reaches 10^{-3} A/cm² for the high beam intensity. This augmentation is related to the correlation between photo absorption and photo generation phenomenon at the bulk of β -Ga₂O₃ layer across the IZTO contact and the resulted photocurrent. This outcome concurs with the ones that was attained in previous works (Labeled et al. 2022; Periyagounder et al. 2018; Wu et al. 2021).

3.3 Effect of β -Ga₂O₃ layer thickness

Both beam power and optical wavelength are fixed at 10^2 mW/cm² and 255 nm, respectively. The effect of Ga₂O₃ layer thickness on the I–V characteristics is shown in Fig. 4a. It can be noted that the current density increases with increasing Ga₂O₃ layer thickness. The thicker semiconductor layer, leads to an increase of electron–hole pairs generation, which is manifested by an increase in photocurrent (Sharma et al. 2017). Figure 4b shows the linearity of the photocurrent as function of thickness rise.

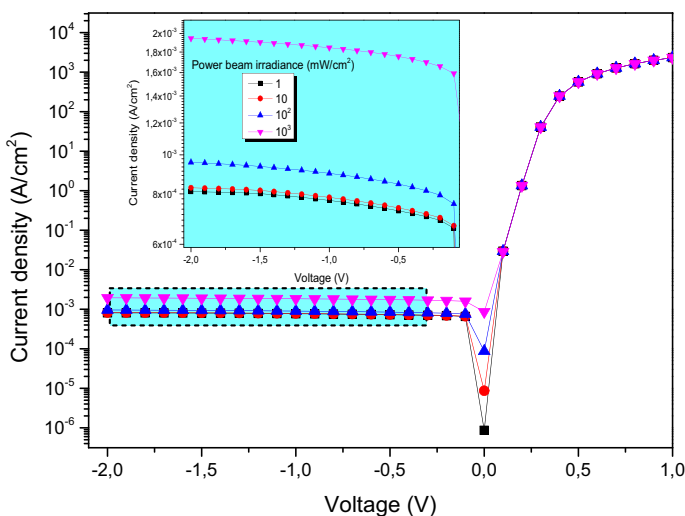


Fig. 3 Simulated I–V characteristics of β -Ga₂O₃ photodiode under different light intensities

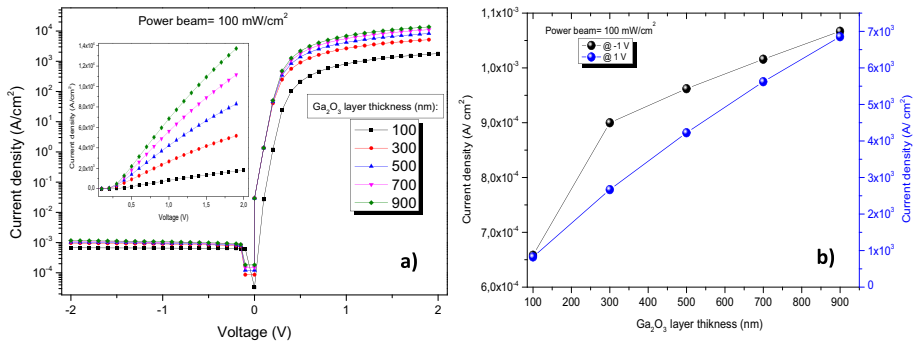


Fig. 4 **a** Simulated I–V characteristics from 100 to 900 nm thickness under an illumination of 100 mW/cm² and **b** Current density as function of layer thickness at –1 and 1 V

3.4 Effect of Schottky contact work function

Another parameter that can affect the I–V characteristic of the device in question is the work function (W_f) variation for the IZTO Schottky electrode (Anode). As can be seen from Fig. 5, for fixed 300 nm thick structure, by varying the W_f from 3.78 to 4.58 eV, an increase in current density was observed with decreasing work function until it saturates at 6.2×10^3 A/cm² for a 3.98 eV work function. This outcome is related to the diminution in the barrier height (ϕ_b) at the heterojunction interface, which leads for carriers charge being able to surmount the lowest barrier height (Filali et al. 2017).

3.5 Effect of Ga₂O₃ affinity

Additional, as shown in Fig. 6 below. At direct bias polarization, an apparent influence of the affinity of β -Ga₂O₃ current–voltage characteristics was remarked. Seeing first the shift

Fig. 5 Simulated I–V characteristics for different metal's work function

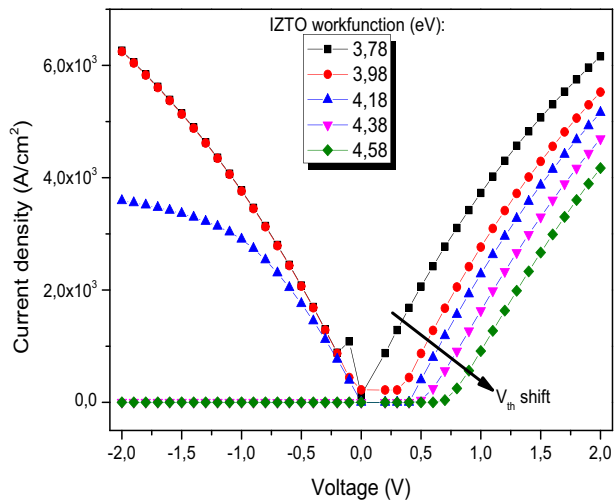
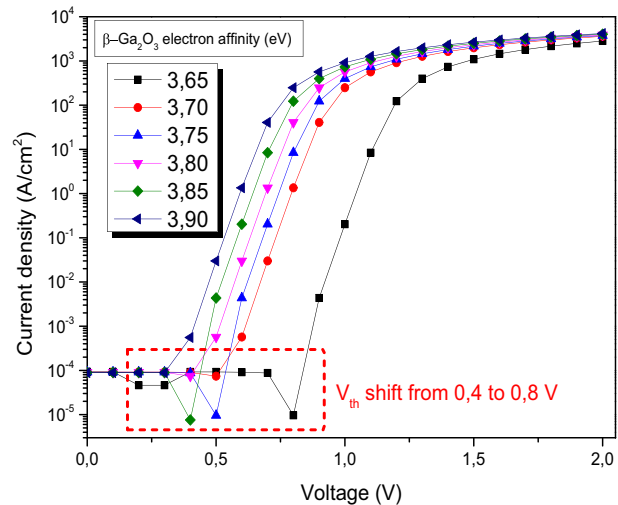


Fig. 6 Simulated I–V characteristics for different semiconductor affinities



of the threshold voltage (V_{th}) from 0.22 V to 0.8 V as well as the insignificant decrease of the saturation current in function of increasing affinity of semiconductors. These results are in good agreement with those obtained in the literature where the affinity is highly dependent to the structural and morphological composition of the semiconductor layer (Labeled et al. 2022; Mao 2007).

3.6 Effect of β -Ga₂O₃ doping concentration

Figure 7a shows the effect of doping concentration variation on the electrical I–V characteristics of the photodetector, for both reverse and forward current density. As the doping concentration increases from 3×10^{16} to 10^{18} cm^{-3} , a significant rise in current density from 10^{-3} to 10^{-2} A/cm^2 is noticed. These phenomena mainly due to the

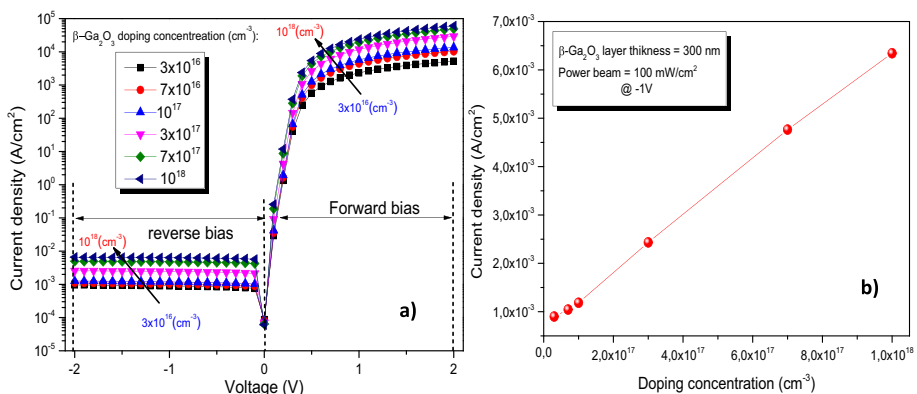


Fig. 7 **a** Simulated I–V characteristics of β -Ga₂O₃ photodiode for different doping concentration and **b** plot of current density as function of β -Ga₂O₃ doping concentration

enhancement in the carriers (electron–hole) within the bulk of $\beta\text{-Ga}_2\text{O}_3$, when they react according to the bias voltage under illumination to participate on the photo-generation mechanism, in addition to the important impact of doping concentration on the difference between conduction band and Fermi level energy which will affect the conductivity of the device (Mao 2007; Sze et al. 2021). As depicted in Fig. 7b, an apparent increase in photocurrent with increasing in semiconductor doping concentration, this is with correlation with result shows in Fig. 7a.

As presented in Fig. 8, an apparent effect of doping concentration of Ga_2O_3 semiconductor layer on the band bending of the conduction and valence energy bands. One can see that heavily doped semiconductor has narrow space-charge region width that allows electrons to tunnel from the metal to semiconductor and from semiconductor to the metal. This phenomenon is called tunneling effect.

3.7 Effect of ohmic contact material

In order to evaluate the performance for different cathode contact materials, four transparent conductive oxides: Aluminium-Zinc Oxide (AZO), Fluorine-Tin oxide (FTO), Indium-Tin oxide (ITO) and Indium-Zinc oxide (IZO) were tested. The choice of those TCO's was based on their both electrical and optical properties discussed in several previous works (Choi et al. 2010; Heo et al. 2013). Figure 9 shows the current density–voltage (J–V) at the forward bias. Knowing that both Ga_2O_3 layer thickness and beam power intensity are fixed to 300 nm and 10^2 mW/cm^2 , respectively. It can be seen that there is large gap between the current values of ITO and AZO contacts compared to that of IZO and FTO. This difference is around 4 decades till reaches $5 \times 10^3 \text{ A/cm}^2$. This difference is commonly attributed to the good but not perfect transmission properties of the used TCO's that leads to some absorption losses in photodetection. On the other hand, materials stoichiometry and defects in bulk could affect the resistivity, which lead to an increase or decrease on electrical contact conductivity (Chesson 2021; Choi et al. 2010; Fang et al. 2006).

Fig. 8 Equilibrium bands diagram variation with $\beta\text{-Ga}_2\text{O}_3$ doping concentration change at 0.255 μm optical wavelength and 10^2 mW/cm^2

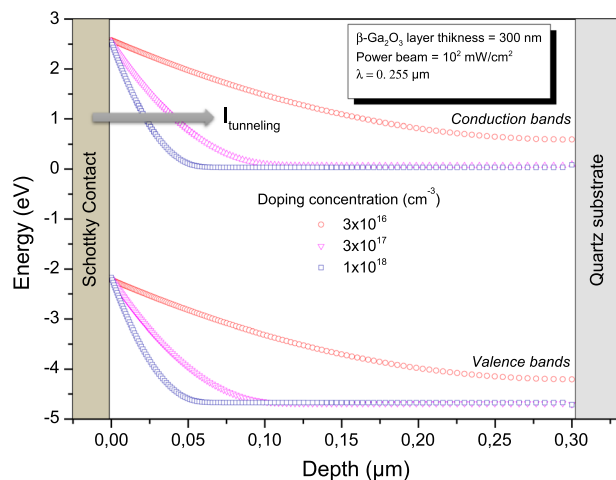
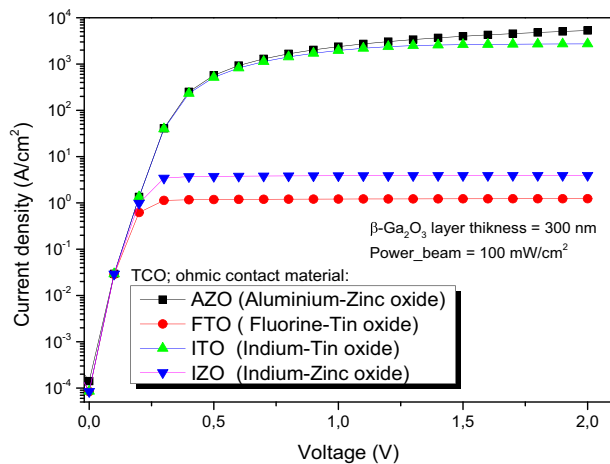


Fig. 9 Simulated I–V characteristics for transparent conductive oxides (TCOs) according to their work function values



3.8 Effect of optical wavelength

In order to investigate the behavior of the Schottky β -Ga₂O₃ based photodetector with different optical wavelengths (UV and visible), the semiconductor absorption layer (Ga₂O₃) thickness is fixed at 300 nm while the beam power is set to 100 mW/cm². The reverse current density as a function of wavelength and responsivity are presented in Fig. 10a, b, respectively. Figure 10c presents the photocurrent versus wavelength variation at -2 V. The current steadily increases with increasing wavelength starting from just above 100 nm and peaks at 250, 254 and 256 nm for 100, 500 and 1000 mW/cm², respectively, the response peak at 255 nm wavelength demonstrates the excellent selectivity of Ga₂O₃ absorption layer, which responds solely to the solar-blind UV with wavelengths below 261 nm. These finding agree with those reported in a previous work (Li et al. 2023; Sharma et al. 2017). Responsivity is another important figure of merit that confirms the well-functioning mechanism of the UV photodetector with specific wavelength. The responsivity (R) (Liao 2022; Sze et al. 2021) is considered as a measure of the effectiveness of the conversion of the light power into electrical current. Hence, R is defined as the ratio of the photocurrent (I_{ph}) to the incident light power (P_{opt}) at a given wavelength:

$$R = \frac{I_{ph}}{P_{opt}} \quad (3)$$

As can be seen from Fig. 11b, the device has shown a maximum responsivity around 0.23 A/W for UV light wavelengths 250, 254 and 256 nm. This outcome is in good agreement with literature bibliography about Ga₂O₃ optical properties, where the wavelengths around 250 nm corresponds to the optical absorption edge of this material (Pearton et al. 2018; Rebien et al. 2002). In addition, these results indicate that the photo-generated carriers in the device are mostly excited by solar-blind radiation (UV light) with a higher energy than Ga₂O₃ band gap, possibly, the deep defect levels, which are associated with oxygen vacancies, are the source of this occurrence. Photocurrent is produced when electrons in oxygen vacancies are stimulated to the conduction band (Labed et al. 2022; Mao 2007; Rebien et al. 2002).

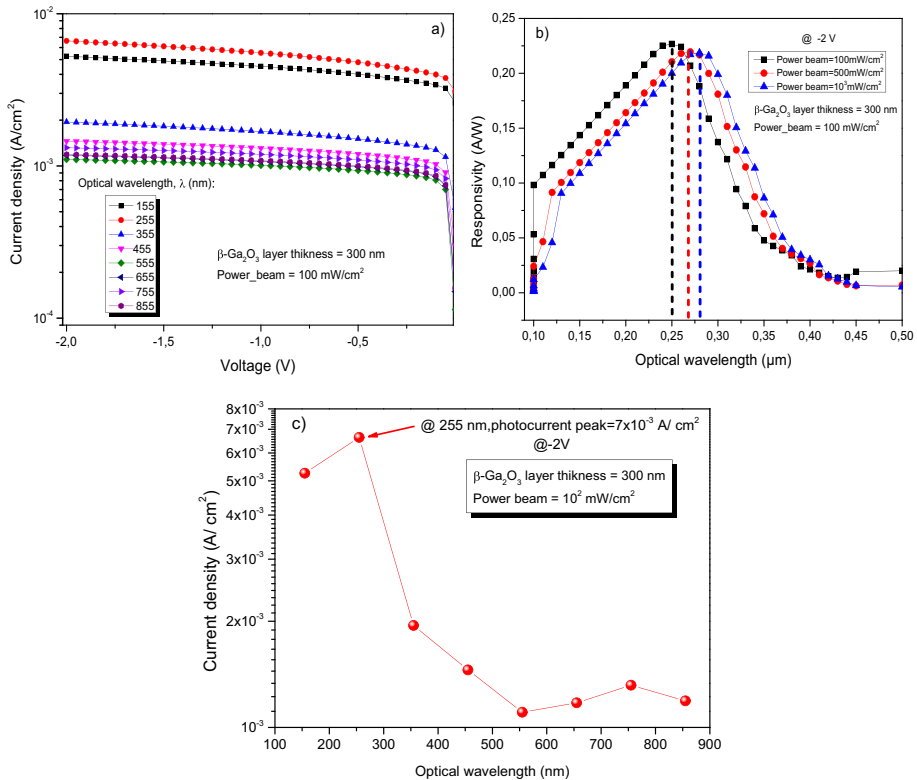


Fig. 10 Simulated **a** I–V characteristics, **b** responsivity as a function of the incident light wavelength and **c** current density as function of optical wavelength variation from 155 to 855 nm at -2 V

Figure 11a–c presents the plot of the photoabsorption rate across Ga₂O₃ layer with different wavelengths irradiance. It can be seen that the high photoabsorption rate value was obtained where we irradiate the photodetector with $0.255 \mu m$ wavelength compared to 0.355 and $0.455 \mu m$, which is in good correlation with result obtained in Fig. 10b (responsivity). Other simulations were done to confirm previous results as depicted in Fig. 11d, the photogeneration rate has the highest value $10^{22} cm^{-3} s^{-1}$ for irradiance of $0.255 \mu m$. A decrease in the photogeneration rate going deeply in the bulk of Ga₂O₃ Layer can be explained by the photons absorption (at $0.255 \mu m$) which have energy higher than E_g of Ga₂O₃, consequently an important carriers charge generation which lead to high measured photocurrent value.

Depending to the used device type, structure and configuration, the detectivity (D) expressed by $(cmHz^{1/2}/W)$ or (Jones) is a pivotal metric used to assess the performance of photodetectors (Chakrabarti et al. 2003; Hazra et al. 2014). It exhibits the capability of the device to describe the slightest detectable optical signals.

$$D = \frac{q\lambda\eta}{hc} \times \left(\frac{R_A}{4kT} \right)^{1/2} \quad (4)$$

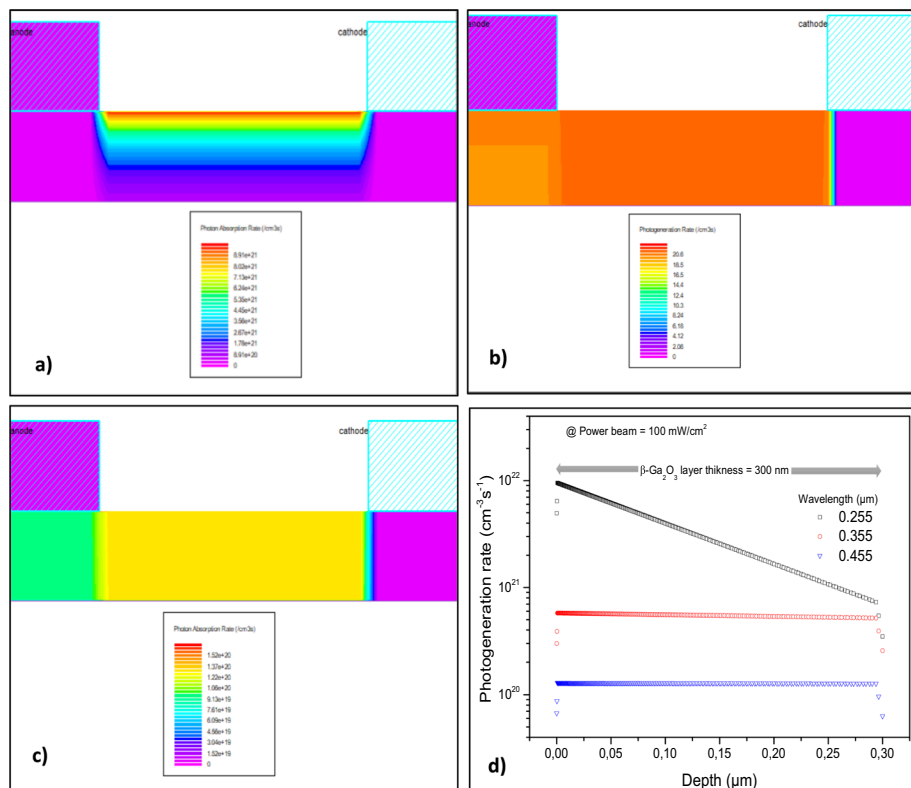


Fig. 11 A Tonyplot of photoabsorption rate at **a** 0.255 μm , **b** 0.355 μm , **c** 0.455 μm and **d** photogeneration rate of the three wavelengths 0.255, 0.355 and 0.455 μm through Ga_2O_3 bulk

where k , T , q , h , c , λ , and η represent Boltzmann constant, lattice temperature, electronic charge, Planck's constant, velocity of light, illumination wavelength, and quantum efficiency [$\eta = R \times (hc/q\lambda)$], respectively. R_A is the resistance-area product of the device and can be obtained from J–V characteristics as follows (Lee et al. 2004):

$$R_A = \left(\frac{\partial J}{\partial V} \right)^{-1} \quad (5)$$

As can be seen from Fig. 12, detectivity value was calculated according to (Ali et al. 2010; Chakrabarti et al. 2003) and found to be 2.5×10^9 (Jones) which is in good accordance with that obtained in a previous work (Wang et al. 2020). Table 2 presents a comparison of some parameters of the present Schottky $\beta\text{-Ga}_2\text{O}_3$ based photodetector with previous works.

4 Conclusion

In this work, a conception and simulation of planar top contacts heterojunction Schottky photodetector based on β -phase gallium oxide semiconductor were investigated. The design and simulation phases were carried out using SilvacoTM TCAD simulation software.

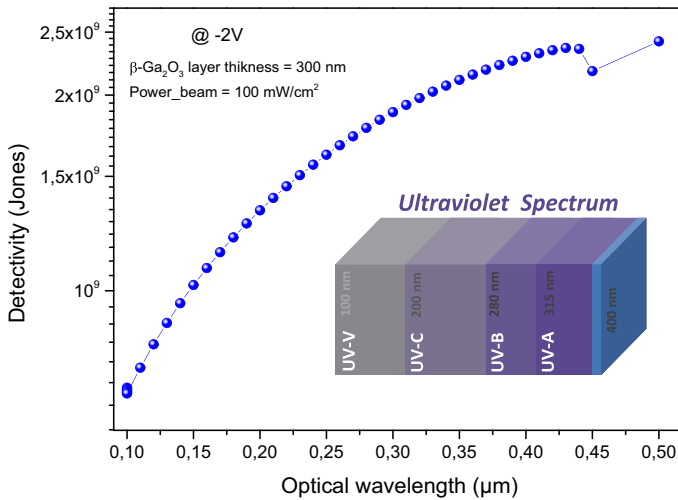


Fig. 12 Photodetector detectivity as a function of optical wavelength for -2 V bias voltage and 100 mW/cm² power light intensity

Table 2 presents a comparison of some parameters of the present Schottky $\beta\text{-Ga}_2\text{O}_3$ based photodetector with previous works

substrate	Contacts / semiconductor	Bias (V)	$I_{\text{Photo}} / I_{\text{Dark}}$	R (mA/W)	D (Jones)	Ref
Diamond	Ti/Au/ $\beta\text{-Ga}_2\text{O}_3$	0	37	0.2	6.9×10^9	Chen et al. (2018)
Sapphire	Ag/ Ga_2O_3	–	122	5.7×10^{-2}	5.4×10^9	Wang et al. (2020)
Quartz	IZTO/ $\beta\text{-Ga}_2\text{O}_3$	-2 V	3×10^4	225	2.5×10^9	Our work
NSTO	Au/Ti/ Ga_2O_3	0	20	2.6	–	Daoyou et al. (2017)
6H-SiC	Au/ $\beta\text{-Ga}_2\text{O}_3$	-2	10^3	68	–	Nakagomi et al. (2013)

The establishment of new materials library for Ga_2O_3 , IZTO and different TCOs with their different properties was incorporated. The simulation results showed that different parameters of $\beta\text{-Ga}_2\text{O}_3$ such as doping concentration, layer thickness, workfunction and electron affinity as well as light power intensity and wavelength affect its opto-electrical performances. These results present a good accordance with several recent published works in terms of rectifying behavior of the Schottky-based photodetector, under dark and UV light irradiance. A contrast ratio found 3×10^4 which signifies the well functioning of the proposed device under dark and illumination conditions. In addition, both responsivity and detectivity were found to be $R=0.23$ A/W and $D=2.5 \times 10^9$ Jones, respectively, which are similar to those obtained in quasi-similar studies. A great dependence on the affinity and workfunction was remarked by the shift in V_{th} values, this prove the correlation between physical phenomenon within the device and the electrical conductivity. At zero bias voltage, a current value of 10^{-3} A/cm² for 100 mW/cm² was observed, which reflect the property of self-powered for the studied photodetector. The entire obtained results can broaden the elaboration of such device and enhance its performances to become an alternative of other Schottky diodes for photodetection.

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Declarations

Conflict of interest The authors declare no competing interests.

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